

Introduction to Data Science and Engineering

- Data visualization in practice



Some materials courtesy of
Rafael A. Irizarry, and are modified
from the original version.

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Slide deck originally created by RB Luo



In this lecture

- Case study: fertility rate
- Case study: income



In this lecture

- **Case study: fertility rate**
- Case study: income

Case study: new insights on poverty

The Gapminder Foundation, an organization dedicated to educating the public by using data to dispel common myths about the so-called developing world.

The organization uses data to show how actual trends in health and economics contradict the narratives that emanate from sensationalist media coverage of catastrophes, tragedies, and other unfortunate events.

Journalists and lobbyists tell dramatic stories. That's their job. They tell stories about extraordinary events and unusual people. The piles of dramatic stories pile up in peoples' minds into an over-dramatic worldview and strong negative stress feelings: "The world is getting worse!", "It's we vs. them!", "Other people are strange!", "The population just keeps growing!" and "Nobody cares!"

Hans Rosling
Professor of International Health
Karolinska Institute

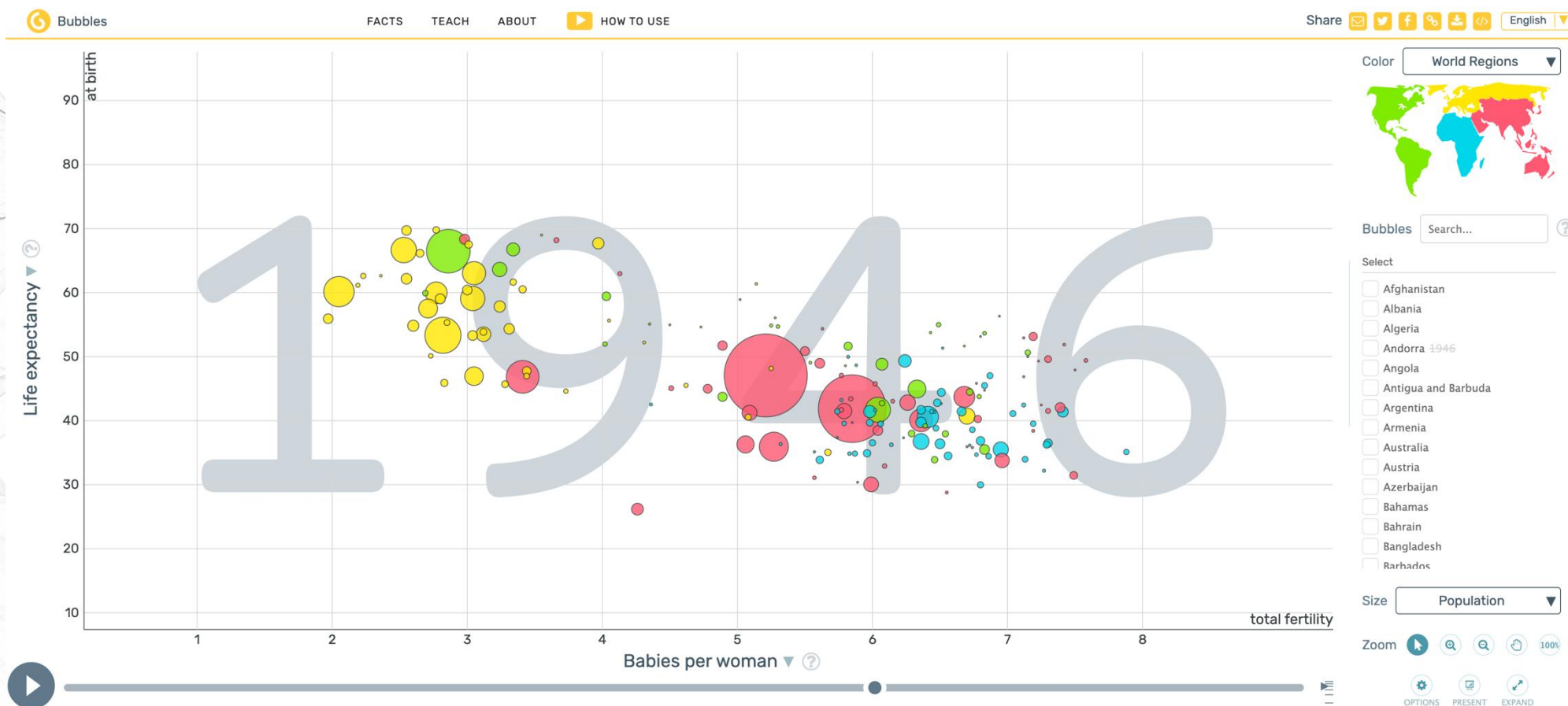


See:

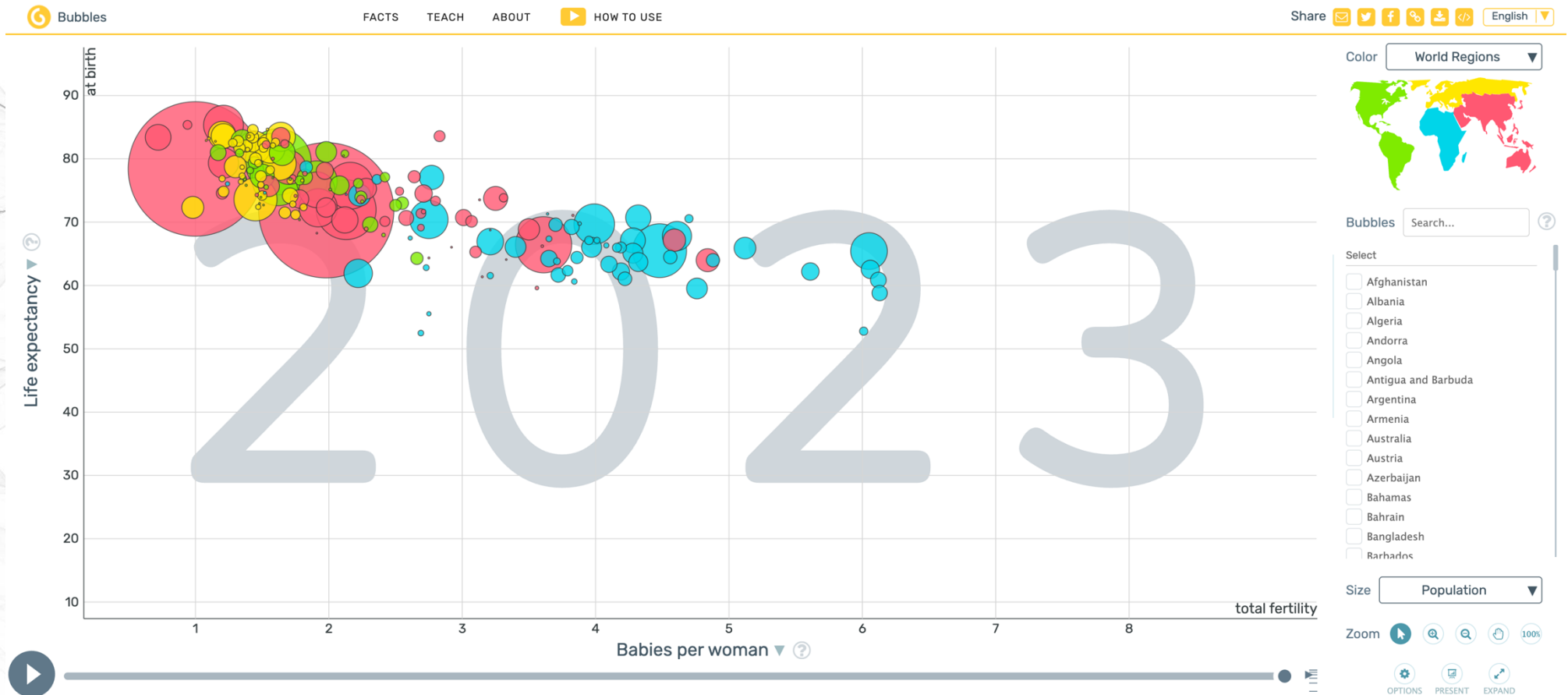
<http://www.gapminder.org/>

https://en.wikipedia.org/wiki/Hans_Rosling

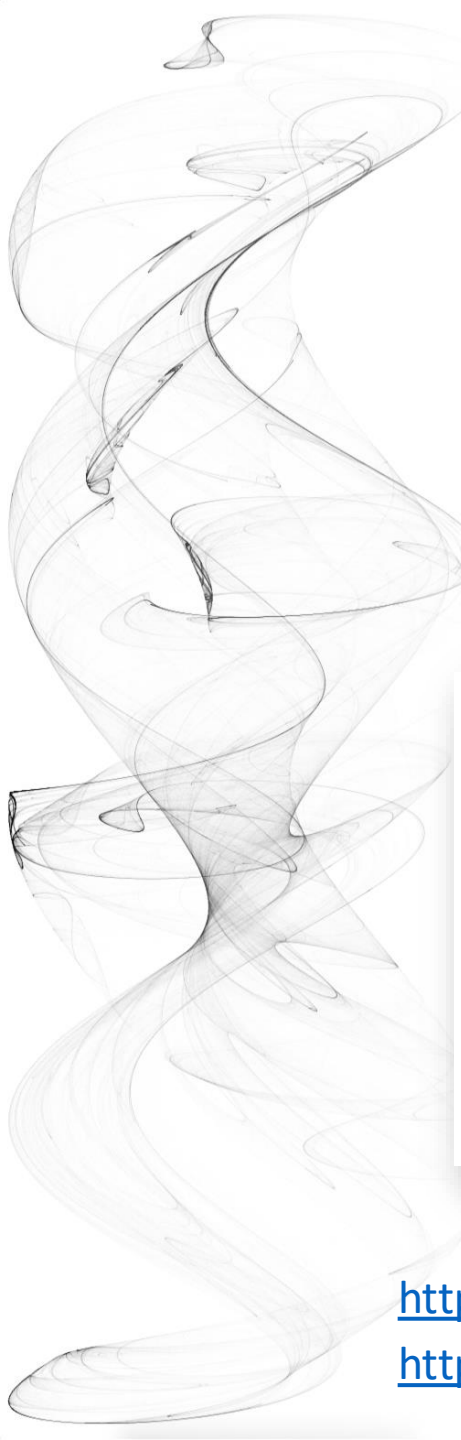
Gapminder's bubble chart



Gapminder's bubble chart



[https://www.gapminder.org/tools/#\\$chart-type=bubbles&url=v2](https://www.gapminder.org/tools/#$chart-type=bubbles&url=v2)



Case study: new insights on poverty

In this case study, we attempt to answer the following question with the help of data: **Is it still a fair characterization of today's world to say it is divided into western nations and the developing world in Africa, Asia, and Latin America?**

```
> library(tidyverse)
> library(dslabs)
> data(gapminder)
> str(gapminder)

'data.frame':  10545 obs. of  9 variables:
 $ country      : Factor w/ 185 levels "Albania","Algeria",...: 1 2 3 4 5 6 7 8 9 10 ...
 $ year         : int  1960 1960 1960 1960 1960 1960 1960 1960 1960 1960 ...
 $ infant_mortality: num  115.4 148.2 208 NA 59.9 ...
 $ life_expectancy: num  62.9 47.5 36 63 65.4 ...
 $ fertility     : num  6.19 7.65 7.32 4.43 3.11 4.55 4.82 3.45 2.7 5.57 ...
 $ population    : num  1636054 11124892 5270844 54681 20619075 ...
 $ gdp           : num  NA 1.38e+10 NA NA 1.08e+11 ...
 $ continent     : Factor w/ 5 levels "Africa","Americas",...: 4 1 1 2 2 3 2 5 4 3 ...
 $ region        : Factor w/ 22 levels "Australia and New Zealand",...: 19 11 10 2 15 21
```

https://www.ted.com/talks/hans_rosling_reveals_new_insights_on_poverty

https://www.ted.com/talks/hans_rosling_shows_the_best_stats_you_ve_ever_seen



Hans Rosling's quiz

We start by testing our knowledge regarding differences in child mortality across different countries.

For each of the six pairs of countries below, which country do you think had the highest child mortality rates in 2015? Which pairs do you think are most similar?

1. Sri Lanka or Turkey
2. Poland or South Korea
3. Malaysia or Russia
4. Pakistan or Vietnam
5. Thailand or South Africa

Try some new questions yourself at: <https://upgrader.gapminder.org/>



```
> gapminder |>
+ filter(year == 2015 & country %in% c("Sri Lanka","Turkey")) |>
+ select(country, infant_mortality)
  country infant_mortality
1 Sri Lanka             8.4
2   Turkey             11.6
```

Turkey has the higher infant mortality rate

country	infant mortality	country	infant mortality
Sri Lanka	8.4	Turkey	11.6
Poland	4.5	South Korea	2.9
Malaysia	6.0	Russia	8.2
Pakistan	65.8	Vietnam	17.3
Thailand	10.5	South Africa	33.6

It turns out that when Hans Rosling gave this quiz to educated groups of people, **the average score was less than 2.5 out of 5**, worse than what they would have obtained had they guessed randomly.



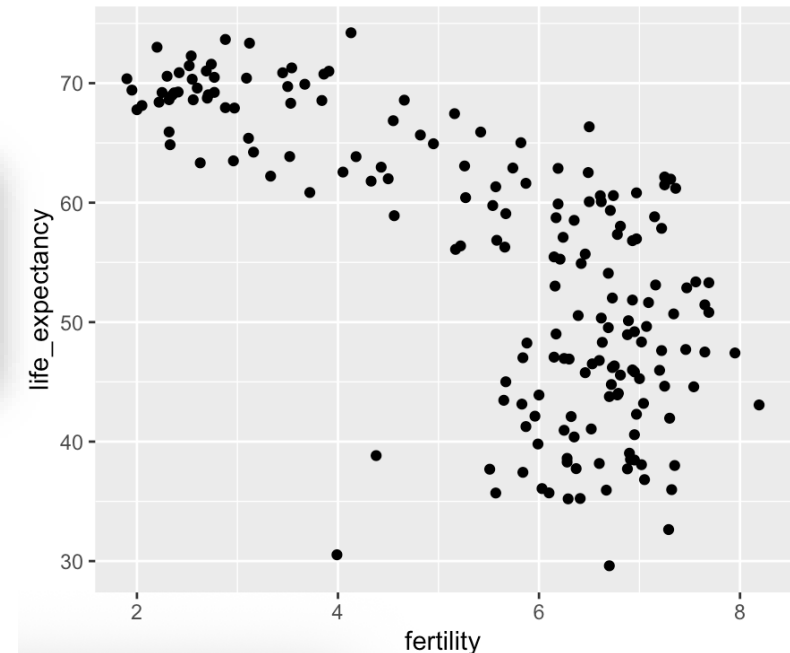
These misconception could stem from the idea that the world is divided into two groups:

- the western world (Western Europe and North America), characterized by longer life spans and small families;
- the developing world (Africa, Asia, and Latin America), characterized by shorter life spans and large families.

Does our data support his conclusion?

We start by plotting life expectancy against fertility rates (average number of children per woman).

```
gapminder |>  
  filter(year == 1960) |>  
  ggplot(aes(fertility, life_expectancy)) +  
  geom_point()
```



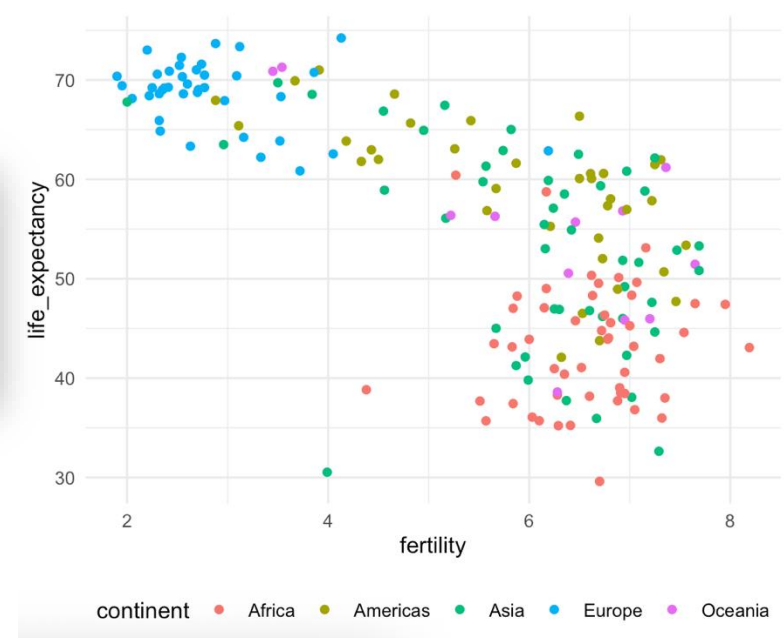
Scatterplots

Most points fall into two distinct categories:

- Life expectancy around 70 years and 3 or fewer children per family
- Life expectancy lower than 65 years and more than 5 children per family

We can further identify each dots using different colors:

```
gapminder |>  
  filter(year == 1960) |>  
  ggplot(aes(fertility, life_expectancy, color = continent)) +  
  geom_point() +  
  theme_minimal() +  
  theme(legend.position = "bottom")
```

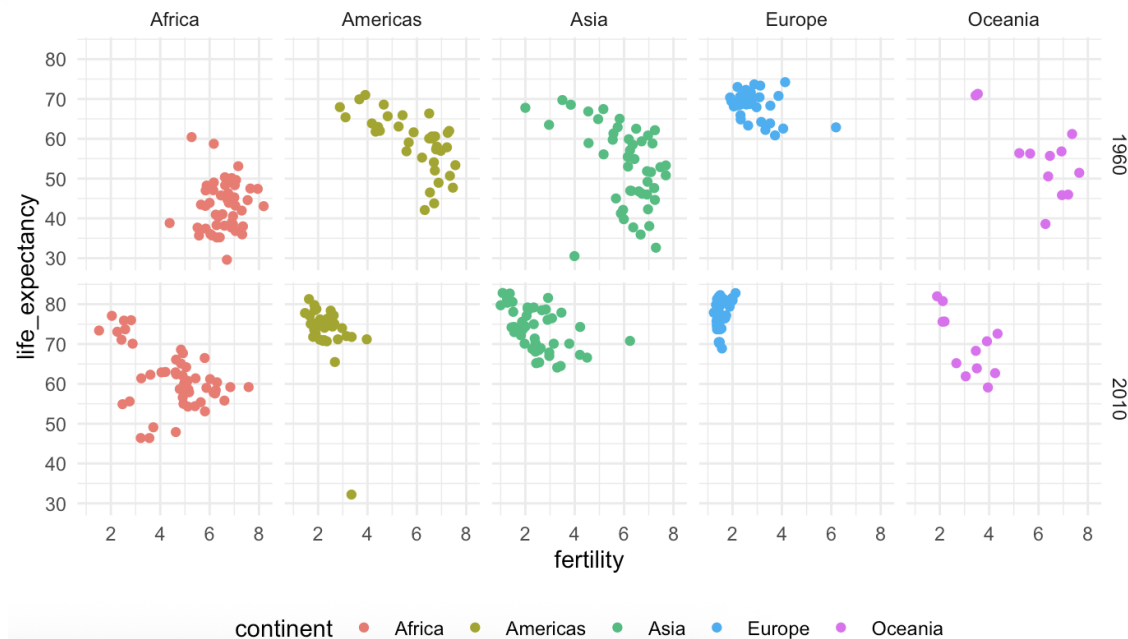


Scatterplots and faceting

Does this dichotomy still exist after 50 years?

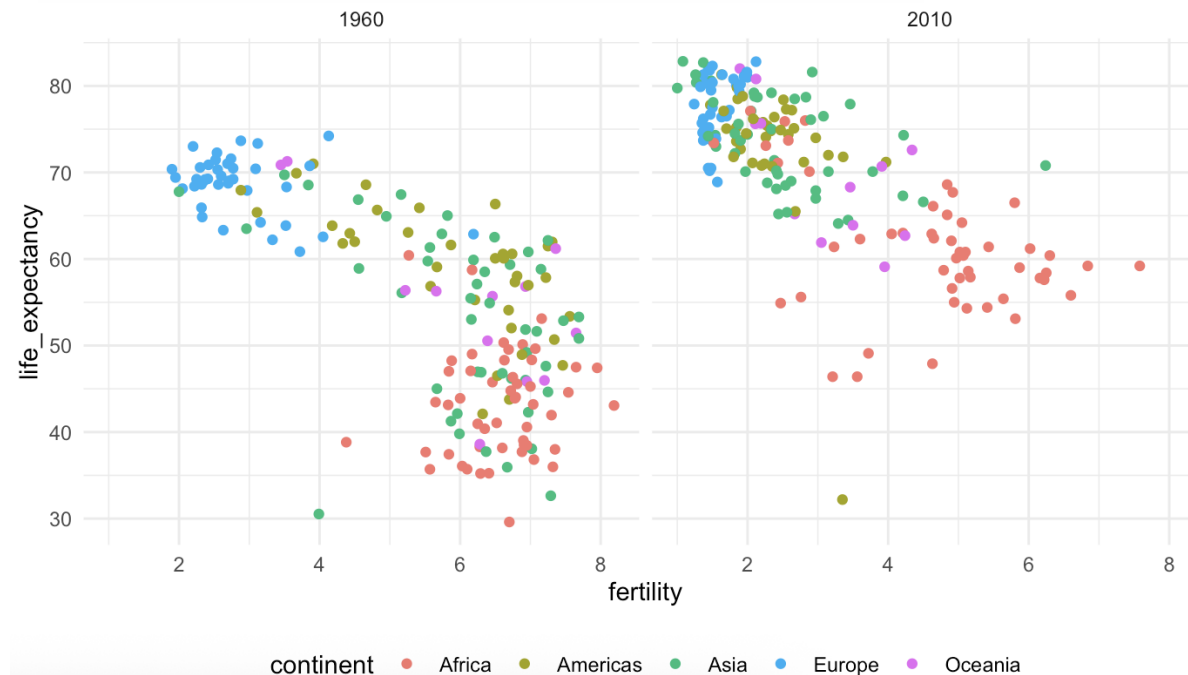
```
gapminder |>
  filter(year %in% c(1960, 2010)) |>
  ggplot(aes(fertility, life_expectancy, color = continent)) +
  geom_point() +
  facet_grid(year~continent) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

We can compare variables side by side by faceting: stratify the data and make the same plot for each strata.



Scatterplots and faceting

```
gapminder |>  
  filter(year %in% c(1960, 2010)) |>  
  ggplot(aes(fertility, life_expectancy, color = continent)) +  
  geom_point() +  
  facet_grid(. ~ year) + Use . to avoid faceting this axis  
  theme_minimal() +  
  theme(legend.position = "bottom")
```



This plot clearly shows that many countries (especially ones in Asia and Americas) have moved from the developing world cluster to the western world one.

The dichotomy view no longer holds in 2010.

Scatterplots and faceting

```
gapminder |>
  filter(year %in% c(1960, 1970, 1980, 1990, 2000, 2010)) |>
  ggplot(aes(fertility, life_expectancy, color = continent)) +
  geom_point() +
  facet_wrap(. ~ year) +
  theme_minimal() +
  theme(legend.position = "bottom")
```



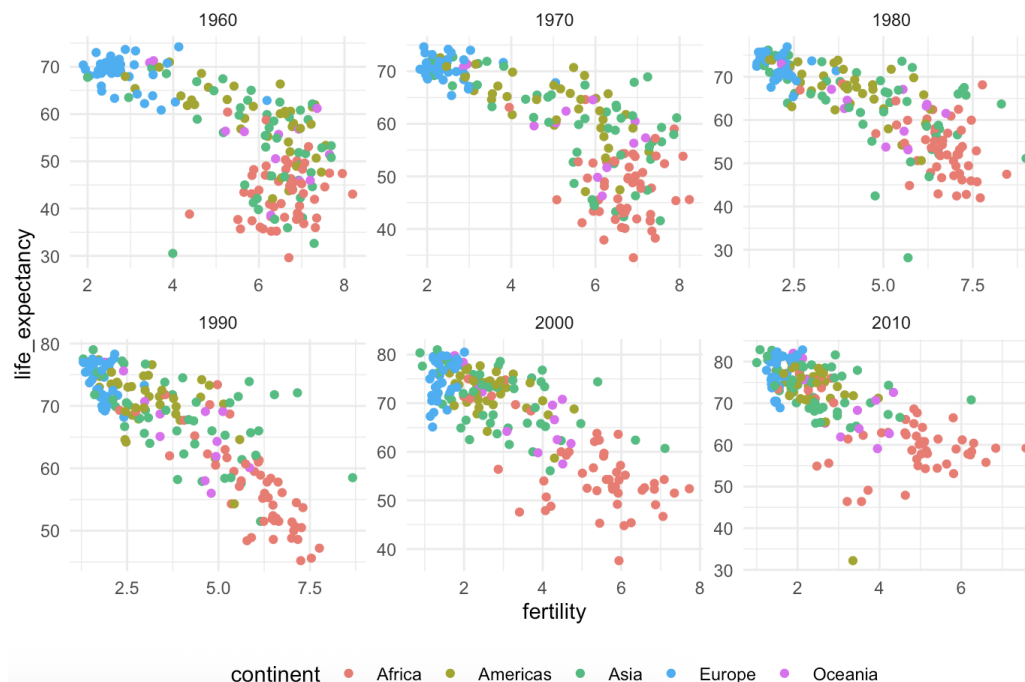
We use the functions `facet_wrap` and `facet_grid` to show the same plot across different strata (defined by different years in this example).

What's their differences?

- Try `?facet_wrap`

Scatterplots and faceting

```
gapminder |>
  filter(year %in% c(1960, 1970, 1980, 1990, 2000, 2010)) |>
  ggplot(aes(fertility, life_expectancy, color = continent)) +
  geom_point() +
  facet_wrap(. ~ year, scales = "free") +
  theme_minimal() +
  theme(legend.position = "bottom")
```



When faceting, the range of data is by default consistent across strata. This makes comparisons easier.

You can change this behavior to scaling within each strata, but this is NOT recommended if the focus is comparison.

Time series plots

Which countries were changing? Were the changes gradual and linear?

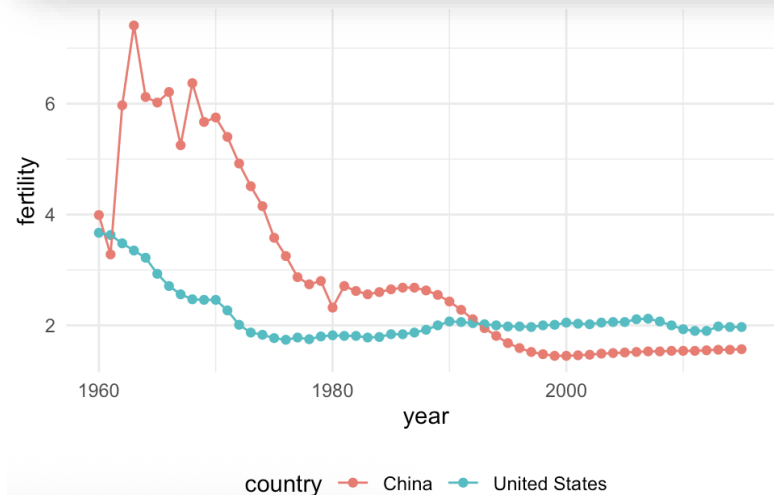
Instead of looking at snapshots, we can learn more by studying the continuous trends over time. Here we introduce time series plots: scatter/line plots with time in the x-axis and an outcome or measurement of interest on the y-axis.

```
> gapminder |>
+   filter(country %in% c("United States", "China")) |>
+   ggplot(aes(year, fertility, color = country)) +
+   geom_point() +
+   geom_line() +
+   theme_minimal() +
+   theme(legend.position = "bottom")
```

Warning messages:

1: Removed 2 rows containing missing values or values outside the scale range (`geom\_point()`).

2: Removed 2 rows containing missing values or values outside the scale range (`geom\_line()`).

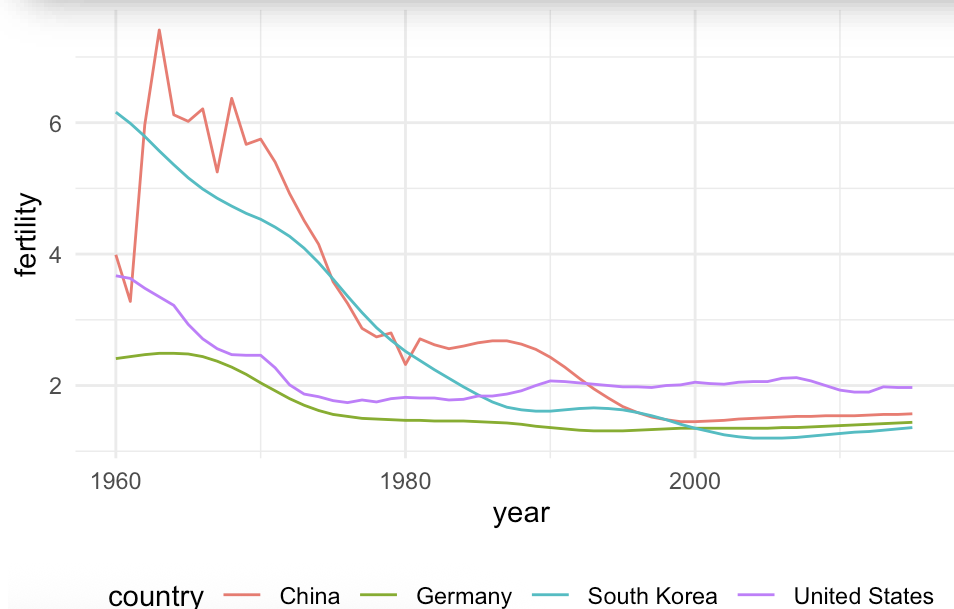


We see that the trend is not linear at all.

- In U.S., there are sharp drop during the 60s and the 70s. Fertility rate later stabilizes around 2.
- In China, family planning policy has a major impact.

More comparisons

```
countries <- c("United States", "China", "Germany", "South Korea")
gapminder |>
  filter(country %in% countries) |>
  ggplot(aes(year, fertility, color = country)) +
  geom_line() +
  theme_minimal() +
  theme(legend.position = "bottom")
```

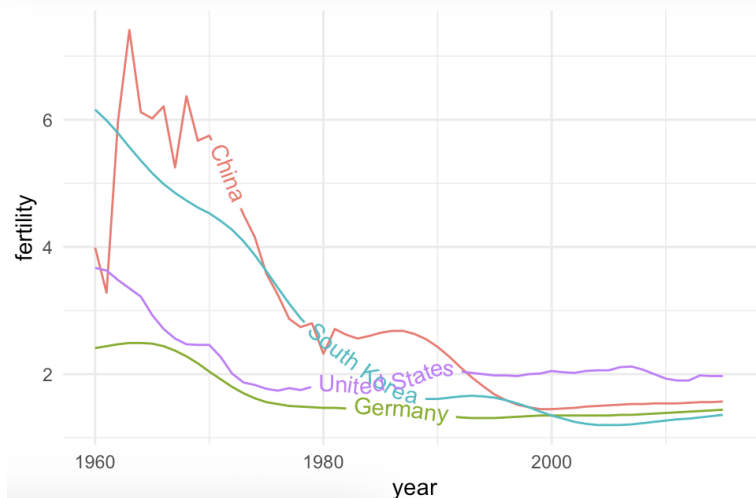


The plot clearly shows how the fertility rates of China and South Korea dropped drastically from 1960 to 2000.

Labels instead of legends

For trend plots we recommend labeling the lines rather than using legends since the viewer can quickly see which line is which country. This suggestion actually applies to most plots: **labeling is usually preferred over legends**. We use the `geom_textpath` package here.

```
countries <- c("United States", "China", "Germany", "South Korea")
gapminder |>
  filter(country %in% countries) |>
  ggplot(aes(year, fertility, color=country, label = country)) +
  geom_textpath() +
  theme_minimal() +
  theme(legend.position = "none")
```



The gap in fertility is almost closed between western and non-western countries



In this lecture

- Case study: fertility rate
- **Case study: income**



Case study: Income

We now shift our attention to the second question related to the commonly held notion that poor countries have become poorer and rich countries become richer:

Has income inequality across countries worsened during the last 40 years?

By using stratification, histograms, smooth densities, and boxplots, we will be able to understand if this is in fact the case.



Case study: Income

The gapminder data table includes a column with the countries' gross domestic product (**GDP**): a measure of the market value of goods and services produced by a country in a year. **GDP per capita** (GDP divided by population) is often used as a rough summary of a country's wealth. Further divided by 365, we have **GDP per capita per day**. Using current US dollars as a unit, a person surviving on **an income of less than \$2.15 a day is defined to be living in absolute poverty** by the World Bank.

```
> gapminder <- gapminder |> mutate(dollars_per_day = gdp/population/365)
```

```
countries_df <- read_csv("https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/countries_df.csv")
colnames(countries_df)[1] <- "geo"
countries_df <- countries_df |> select(geo, name, world_4region, world_6region, west_and_rest)
pop_df <- read_csv("https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/population_df.csv")
gdp_per_capita_df <- read_csv("https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/gdp_per_capita_df.csv")

combined_df <- gdp_per_capita_df

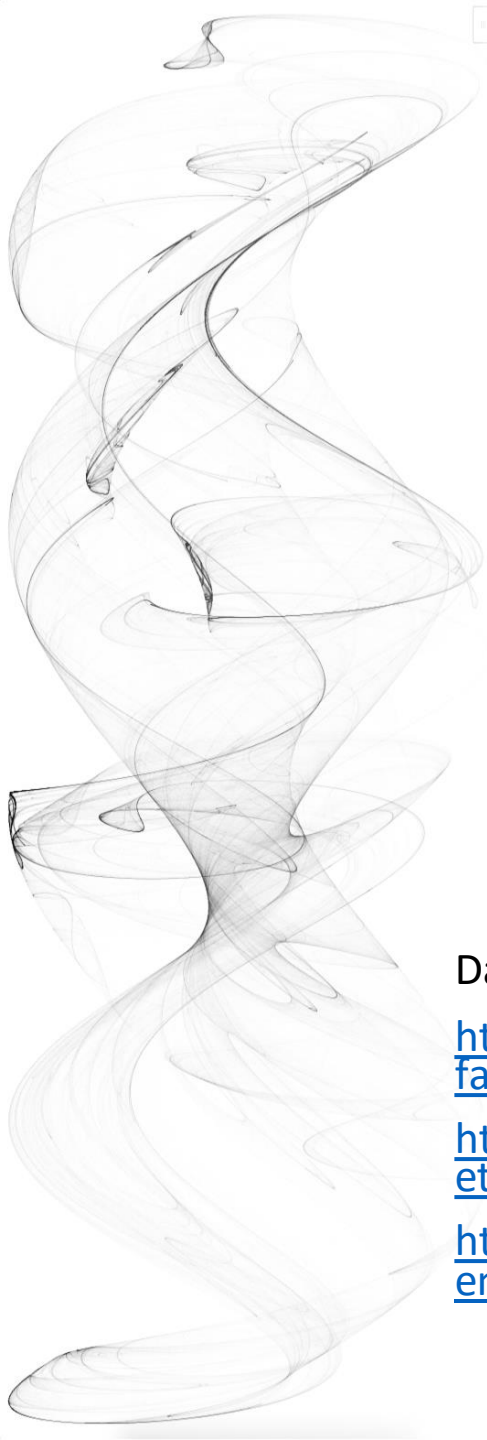
combined_df <- combined_df |>
  filter(between(time, 1950, 2025)) |>
  left_join(pop_df, by = c("geo", "time")) |>
  left_join(countries_df, by = "geo")

country_to_region <- gapminder |>
  select(country, region) |>
  setNames(c("name", "region")) |>
  unique()
combined_df <- combined_df |>
  left_join(country_to_region, by="name")

combined_df <- combined_df |>
  mutate(dollars_per_day = total_gdp_us_inflation_adjusted / total_population_with_projections / 365)
```

See also: <https://www.gapminder.org/data/>

Data source: https://github.com/open-numbers/ddf--gapminder--systema_globalis/tree/master



Case study: Income

```
countries_df <- read_csv("https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/countries_df")
colnames(countries_df)[1] <- "geo"
countries_df <- countries_df |> select(geo, name, world_4region, world_6region, west_and_rest)
pop_df <- read_csv("https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/pop_df")
gdp_per_capita_df <- read_csv("https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/gdp_per_capita_df")

combined_df <- gdp_per_capita_df

combined_df <- combined_df |>
  filter(between(time, 1950, 2025)) |>
  left_join(pop_df, by = c("geo", "time")) |>
  left_join(countries_df, by = "geo")

country_to_region <- gapminder |>
  select(country, region) |>
  setNames(c("name", "region")) |>
  unique()
combined_df <- combined_df |>
  left_join(country_to_region, by="name")

combined_df <- combined_df |>
  mutate(dollars_per_day = total_gdp_us_inflation_adjusted / total_population_with_projections / 365)
```

Data files from:

https://github.com/open-numbers/ddf--gapminder--fasttrack/raw/refs/heads/master/countries_etc_datapoints/ddf--datapoints--pop--by--country--time.csv

https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/countries-etc-datapoints/ddf--datapoints--total_gdp_us_inflation_adjusted--by--geo--time.csv

https://github.com/open-numbers/ddf--gapminder--systema_globalis/raw/refs/heads/master/ddf--entities--geo--country.csv

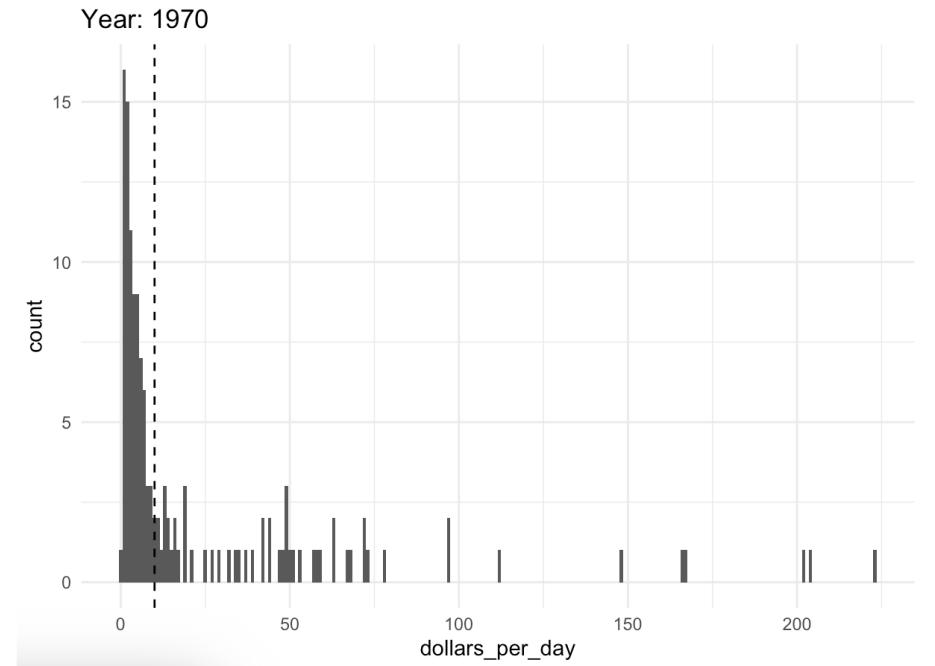
Data transformation

Here is a histogram of per day incomes from 1970

```
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 1) +
  geom_vline(xintercept = 10, lty = 2) +
  theme_minimal() +
  ggtitle("Year: 1970")
```

In this plot, we see that most countries average below \$10 a day (the black dashed line).

Heavy-tailed/skewed: the x-axis expands well beyond \$10 due to the presence of high-income countries.



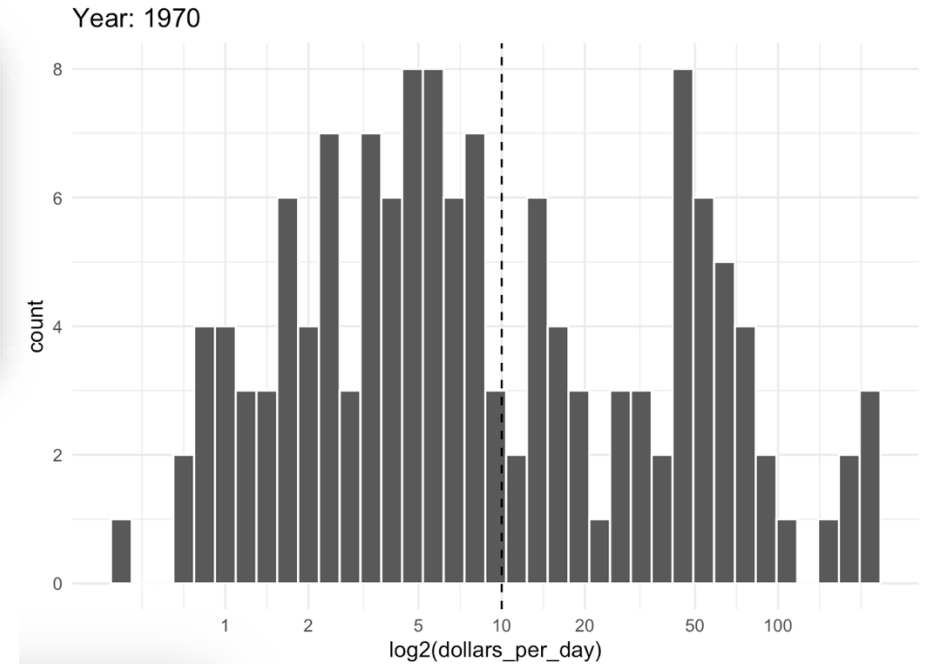
Data transformation

Here is a histogram of per day incomes from 1970

```
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  ggplot(aes(log2(dollars_per_day))) +
  geom_histogram(binwidth = 0.25, color='white') +
  geom_vline(xintercept = log2(10), lty = 2) +
  scale_x_continuous(
    breaks = log2(c(1, 2, 5, 10, 20, 50, 100)), # Log2-transformed scales
    labels = c(1, 2, 5, 10, 20, 50, 100) # Original scales
  ) +
  theme_minimal() +
  ggtitle("Year: 1970")
```

It is very common to use log transformation to deal with heavy-tailed data

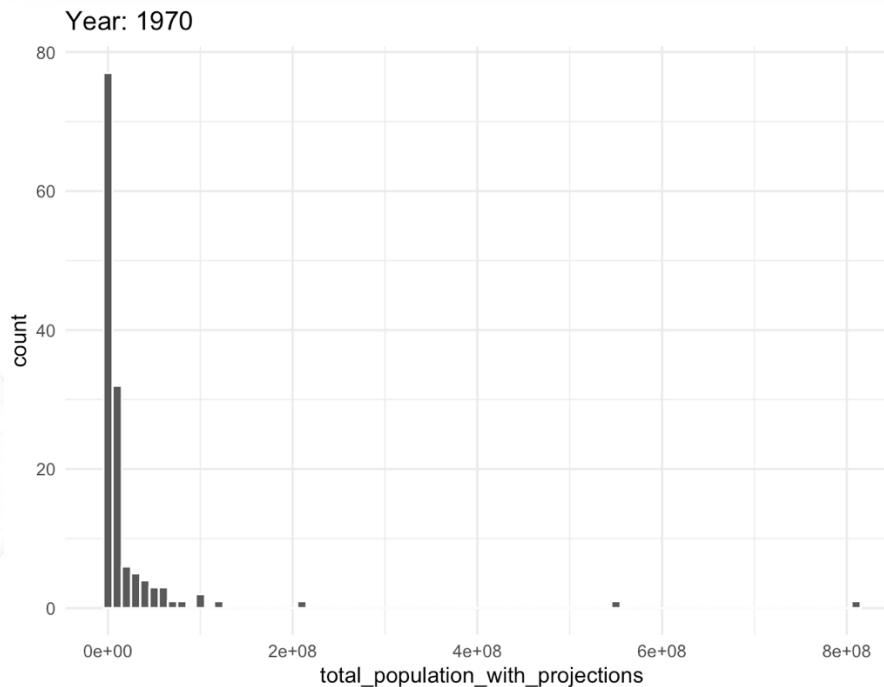
Now it is much easier to see how many countries have average daily incomes of \$1 (extremely poor), \$2 (very poor), \$5 (poor), \$10-20 (middle), \$50 or more (rich).



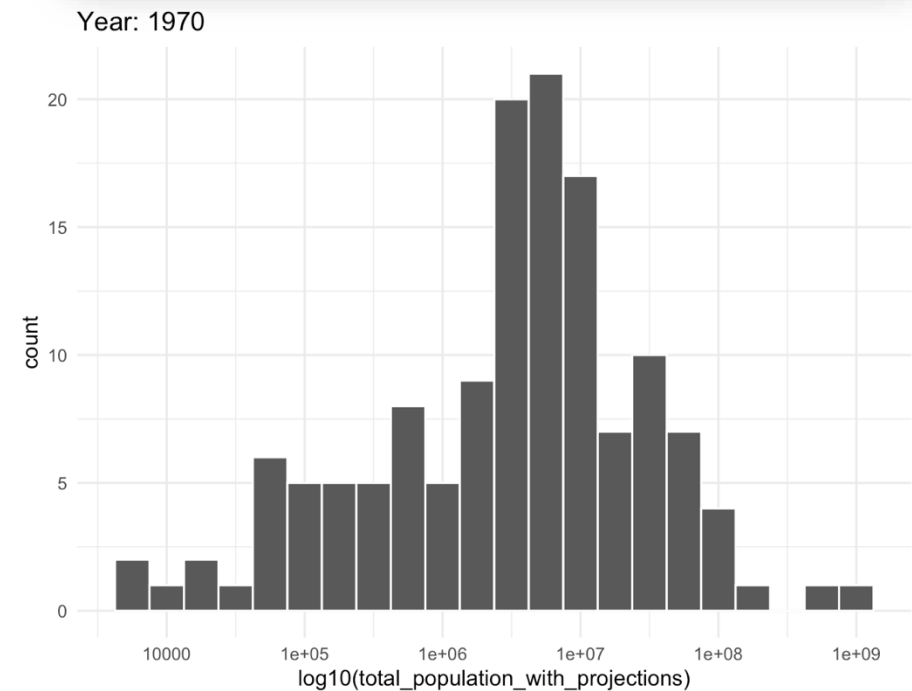
Data transformation

This becomes more obvious when plotting population:

```
combined_df |>
  filter(time == 1970, !is.na(total_population_with_projections)) |>
  ggplot(aes(total_population_with_projections)) +
  geom_histogram(binwidth = 1e7, color='white') +
  theme_minimal() +
  ggtitle("Year: 1970")
```



```
combined_df |>
  filter(time == 1970, !is.na(total_population_with_projections)) |>
  ggplot(aes(log10(total_population_with_projections))) +
  geom_histogram(binwidth = 0.25, color='white') +
  scale_x_continuous(
    breaks = c(4, 5, 6, 7, 8, 9),
    labels = c(1e4, 1e5, 1e6, 1e7, 1e8, 1e9)
  ) +
  theme_minimal() +
  ggtitle("Year: 1970")
```

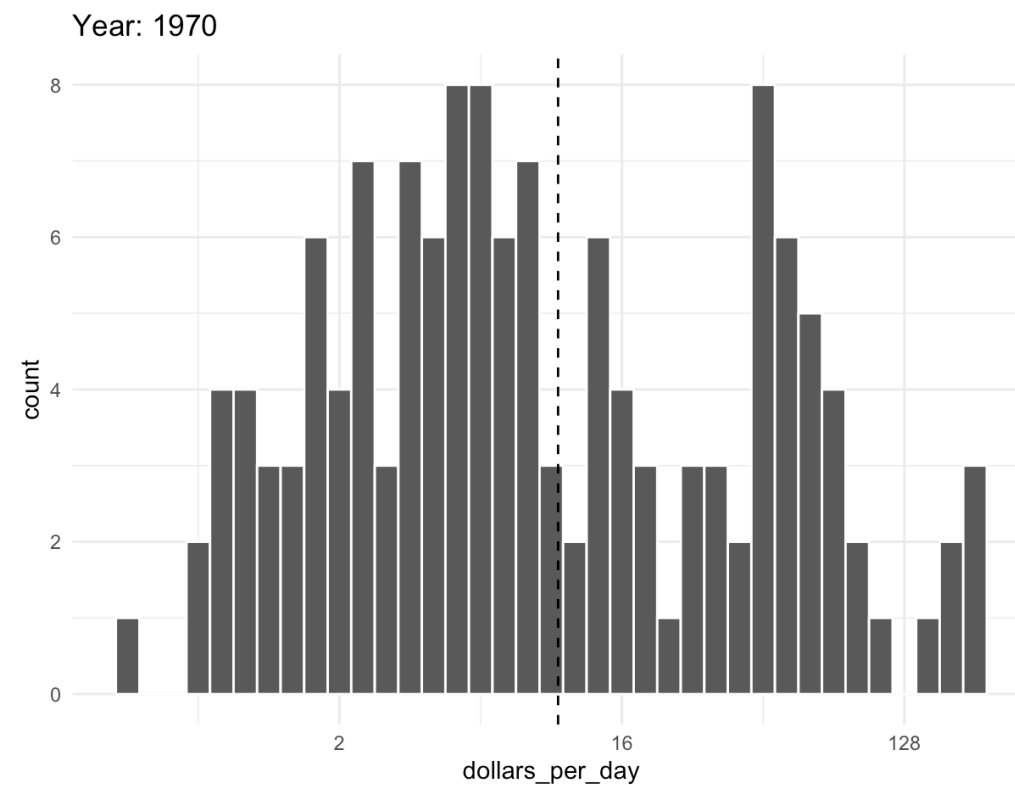


Transform the values or the scale?

Transforming the scale is easier because:

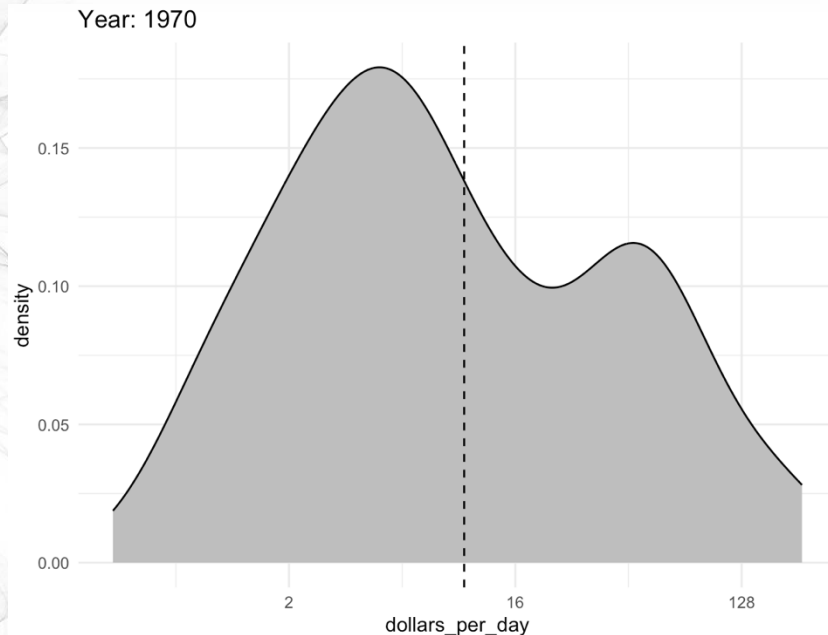
1. The slow speed on transforming values on real large datasets
2. More intuitive axis labels

```
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 0.25, color='white') +
  geom_vline(xintercept = 10, lty = 2) +
  scale_x_continuous(transform="log2") +
  theme_minimal() +
  ggtitle("Year: 1970")
```



Visualizing multimodal distributions

```
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  ggplot(aes(dollars_per_day)) +
  geom_density(fill="grey") +
  geom_vline(xintercept = 10, lty = 2) +
  scale_x_continuous(transform="log2") +
  theme_minimal() +
  ggtitle("Year: 1970")
```



This distribution is not “normal”:

- We see two bumps: one at about 4 and another at about 64. These bumps are often referred to as **modes**. The mode of a distribution is the value with the local maximum frequency.
- Normal distribution has one mode at its average.
- This distribution is bimodal: has two modes.

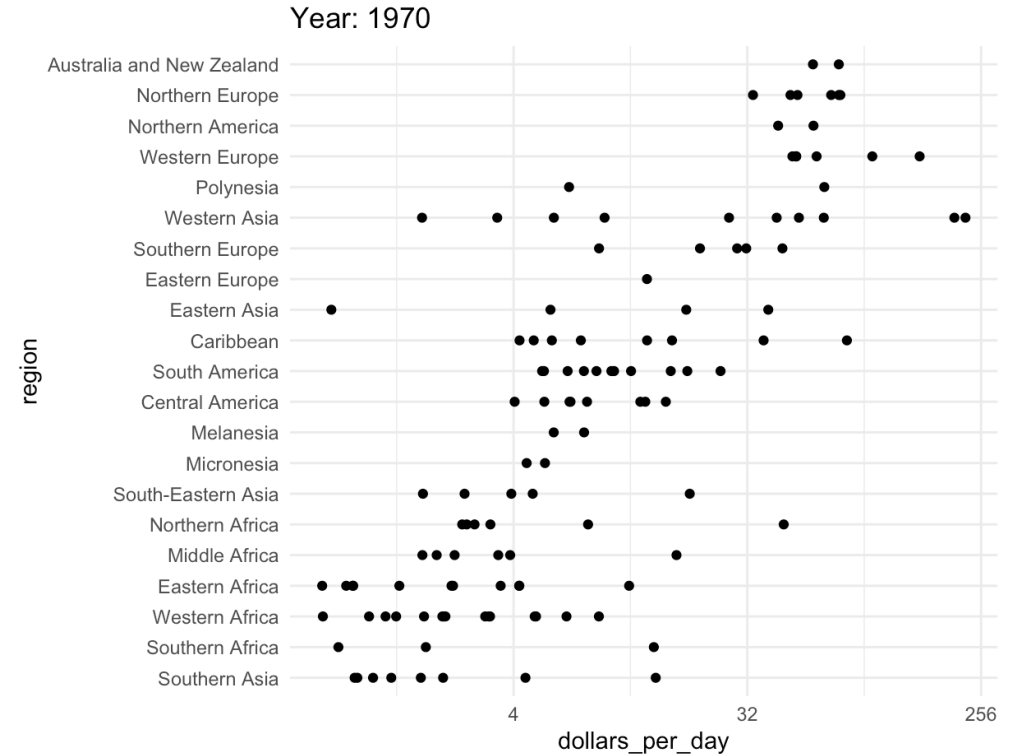
The histogram above suggests that the 1970 country income distribution has two modes: one at about \$2-8/day and another at about \$32-64/day. This bimodality is consistent with a dichotomous world made up of countries with average incomes less than \$10 a day and countries above that

Comparing multiple distributions

The histogram does not show which countries constitute the two modes:

Let's examine daily income by region:

```
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  mutate(region = reorder(region, dollars_per_day, FUN = median)) |>
  filter(!is.na(region)) |>
  ggplot(aes(dollars_per_day, region)) +
  geom_point() +
  scale_x_continuous(transform="log2") +
  theme_minimal() +
  ggtitle("Year: 1970")
```



There exists a “west versus the rest” dichotomy. We see two clear groups, with the rich group composed of North America, Northern and Western Europe, New Zealand and Australia

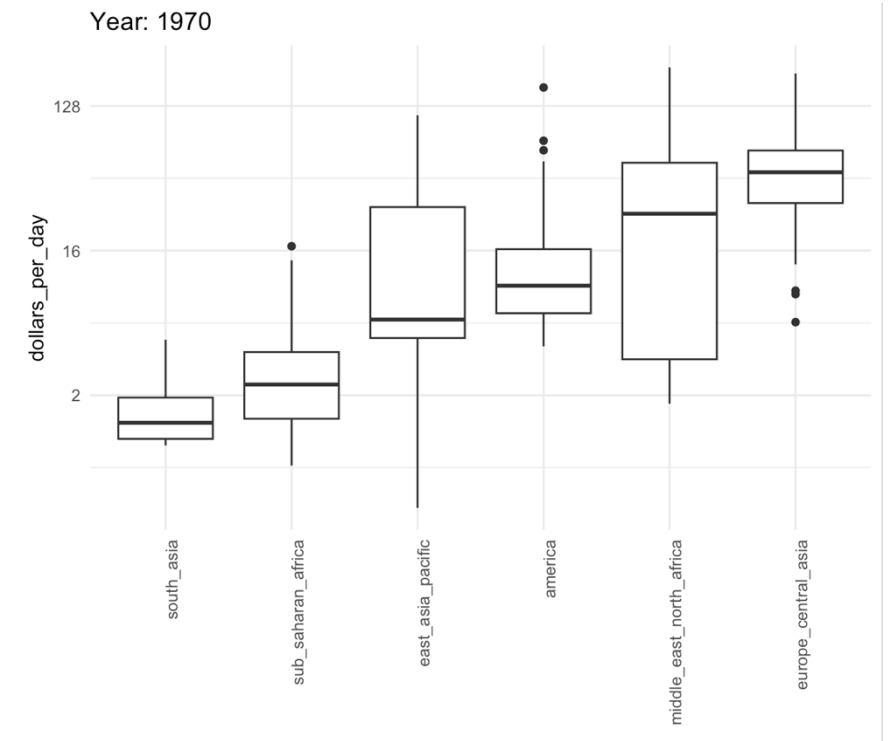
Comparing multiple distributions

The histogram does not show which countries constitute the two modes:

Let's examine daily income by region:

```
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  mutate(world_6region = reorder(world_6region, dollars_per_day, FUN = median)) |>
  ggplot(aes(world_6region, dollars_per_day)) +
  geom_boxplot() +
  scale_y_continuous(transform="log2") +
  xlab("") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle=90, hjust=1)) +
  ggtitle("Year: 1970")
```

Rotate x tick labels by 90 degrees and align them

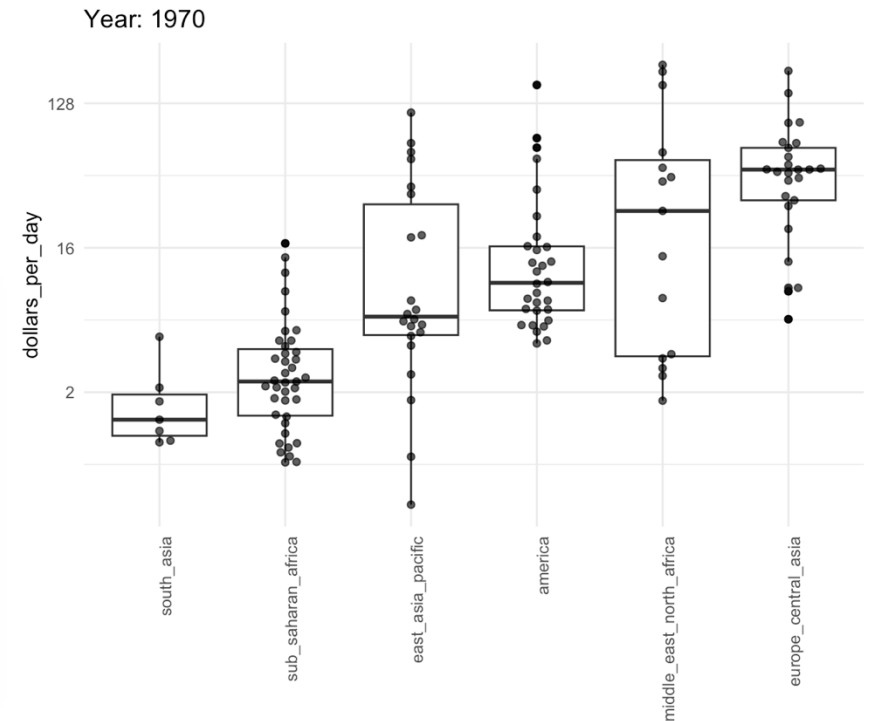


Comparing multiple distributions

Swarm plot (from package `ggbeeswarm`) shows scatterplot between a categorical variable and a continuous variable

```
library(ggbeeswarm)
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  mutate(world_6region = reorder(world_6region, dollars_per_day, FUN = median)) |>
  ggplot(aes(world_6region, dollars_per_day)) +
  geom_boxplot() +
  geom_beeswarm(size = 1.5, cex = 1.5, alpha = 0.7) +
  scale_y_continuous(transform = log2) +
  xlab("") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle=90, hjust=1)) +
  ggtitle("Year: 1970")
```

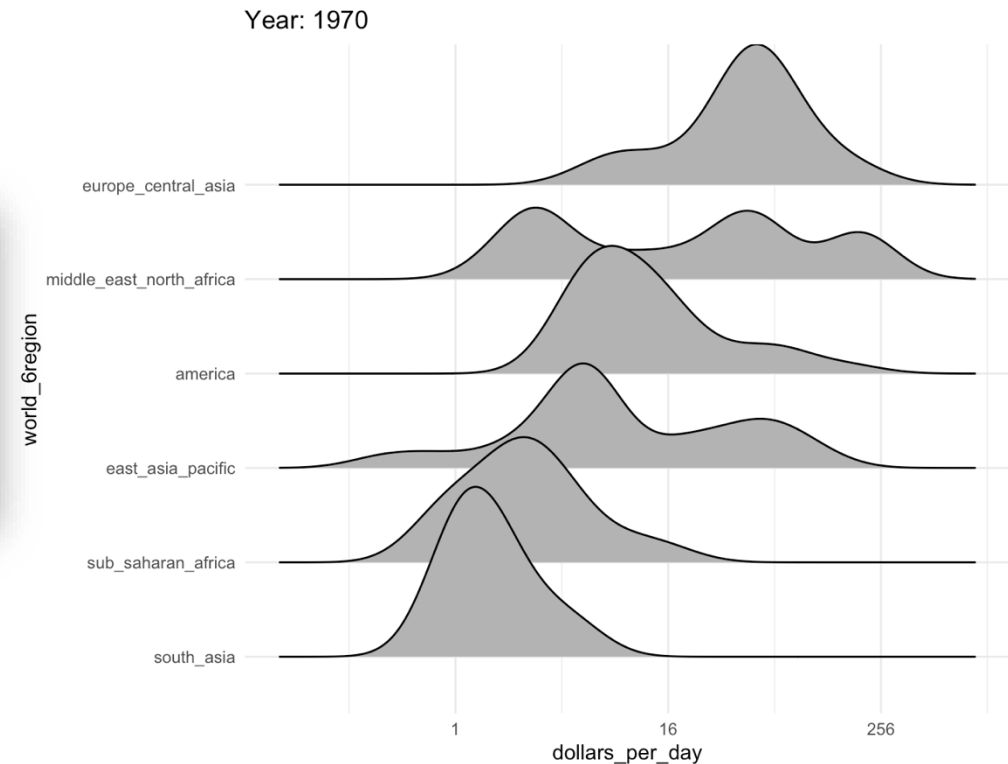
Use scatter plot to show the actual data points



Comparing multiple distributions

Ridge plot (from package `ggridges`) show stacked smooth densities

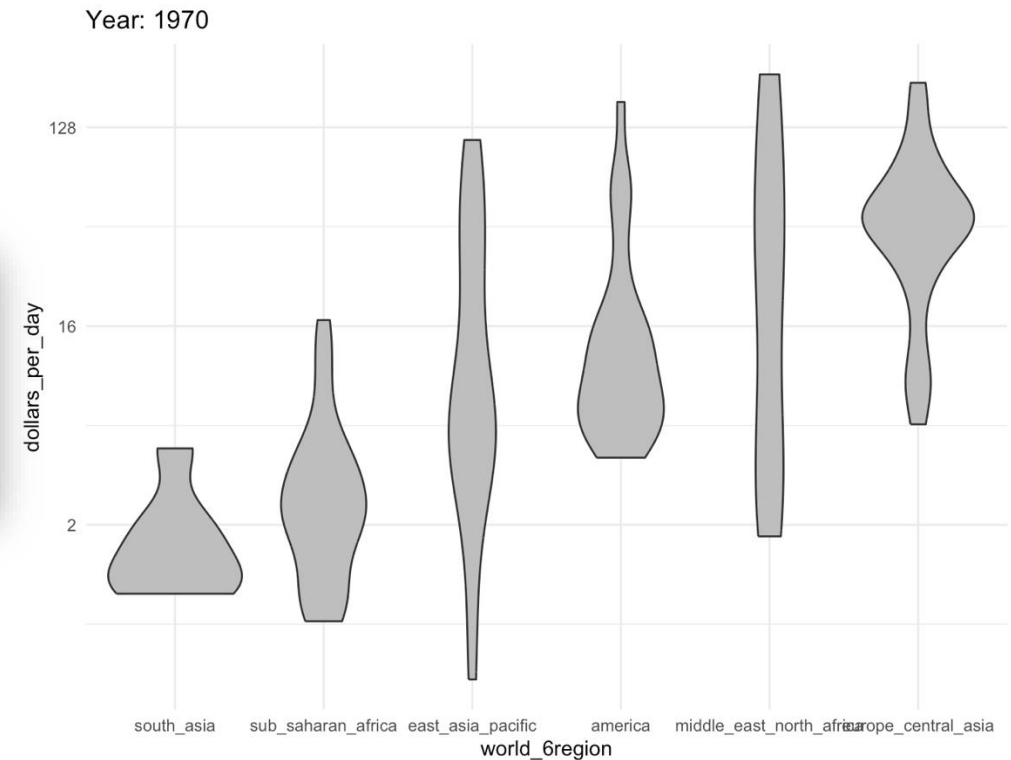
```
library(ggridges)
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  mutate(world_6region = reorder(world_6region, dollars_per_day, FUN = median)) |>
  ggplot(aes(dollars_per_day, world_6region)) +
  geom_density_ridges() +
  scale_x_continuous(transform="log2") +
  theme_minimal() +
  ggtitle("Year: 1970")
```



Comparing multiple distributions

Violin plot is another way of showing multiple smooth densities

```
combined_df |>
  filter(time == 1970, !is.na(dollars_per_day)) |>
  mutate(world_6region = reorder(world_6region, dollars_per_day, FUN = median)) |>
  ggplot(aes(world_6region, dollars_per_day)) +
  geom_violin(fill = "grey") +
  scale_y_continuous(transform="log2") +
  theme_minimal() +
  ggtitle("Year: 1970")
```

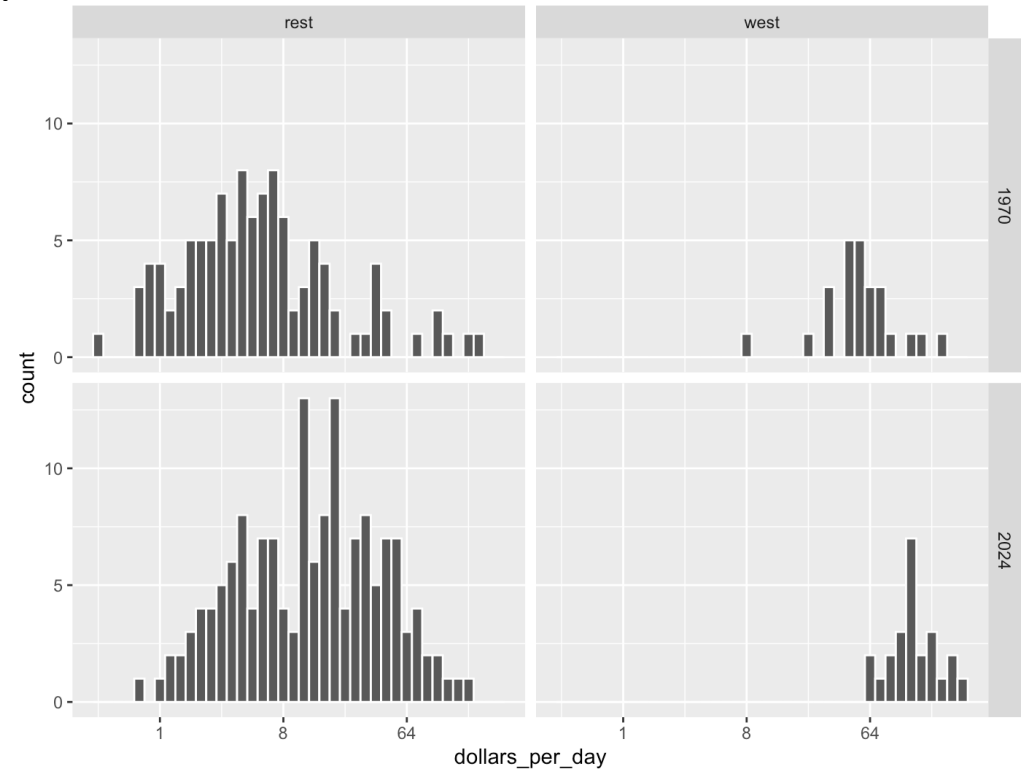


1970 versus 2024

Data exploration clearly shows that in 1970 there was a “west versus the rest” dichotomy. But does this dichotomy persist?

```
past_year <- 1970
present_year <- 2024

combined_df |>
  filter(time %in% c(past_year, present_year), !is.na(west_and_rest)) |>
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 0.25, color = 'white') +
  facet_grid(time ~ west_and_rest) +
  scale_x_continuous(trans = "log2")
```



The plot seems to have more countries in 2024?

1970 versus 2024

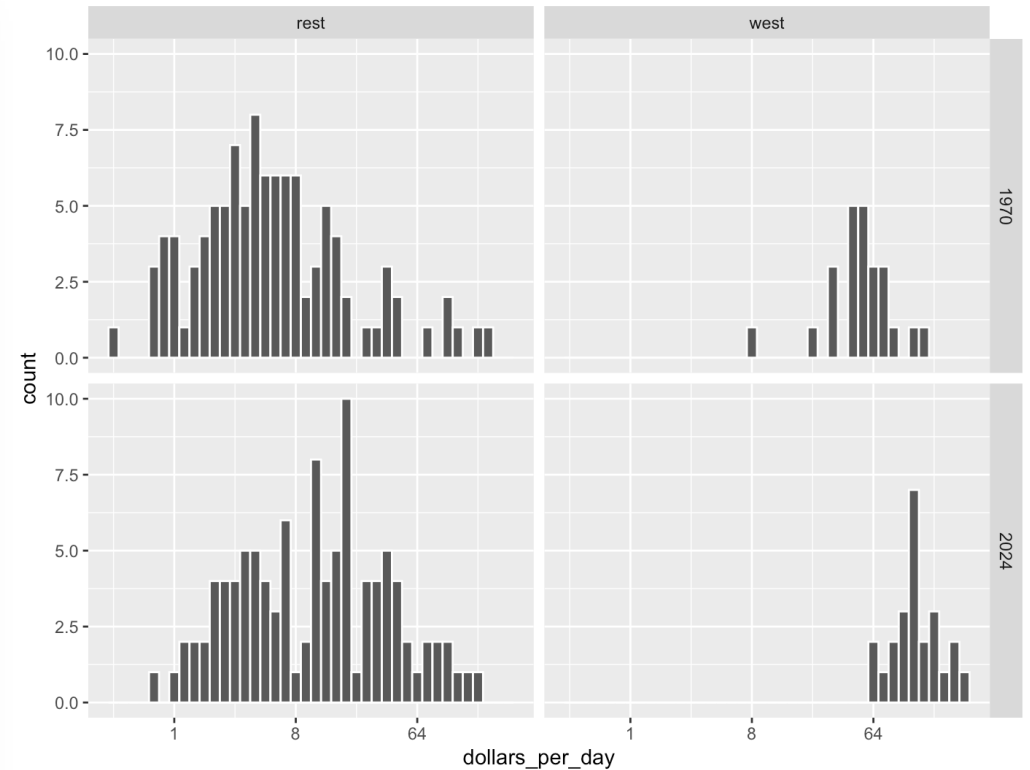
```
past_year <- 1970
present_year <- 2024

country_list1 <- combined_df |>
  filter(time == past_year, !is.na(west_and_rest)) |>
  pull(name)

country_list2 <- combined_df |>
  filter(time == present_year, !is.na(west_and_rest)) |>
  pull(name)

country_list <- intersect(country_list1, country_list2)

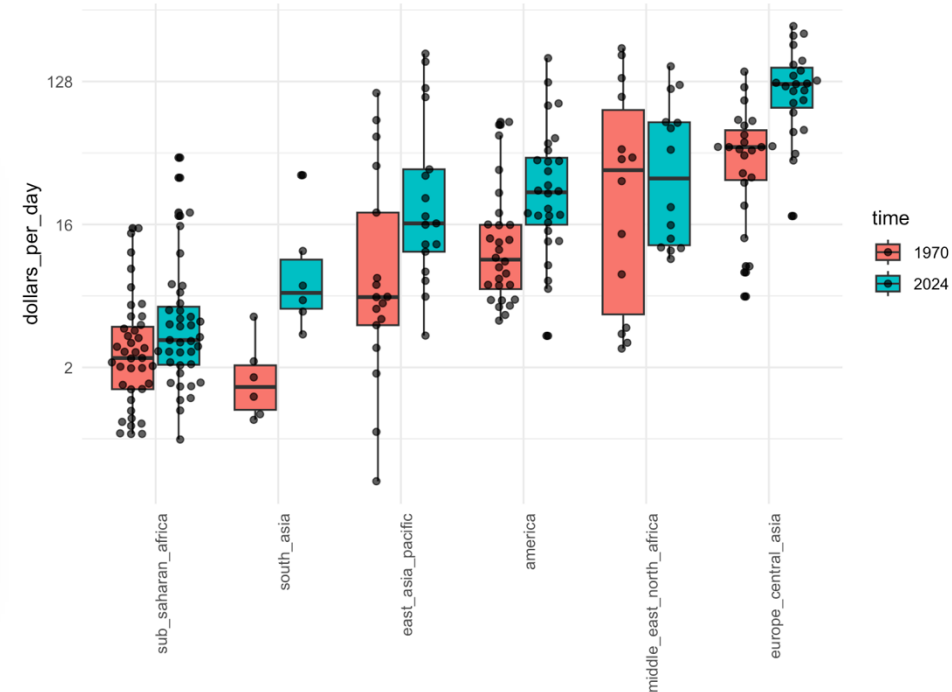
combined_df |>
  filter(time %in% c(past_year, present_year),
         name %in% country_list,
         !is.na(west_and_rest)) |>
  ggplot(aes(dollars_per_day)) +
  geom_histogram(binwidth = 0.25, color = 'white') +
  facet_grid(time ~ west_and_rest) +
  scale_x_continuous(trans = "log2")
```



1970 versus 2024

More conveniently, we can compare the two years side by side by using `aes(fill=year)`. However, `fill` accepts only factor, thus, we convert year into factor after filter

```
combined_df |>
  filter(time %in% c(past_year, present_year),
         name %in% country_list,
         !is.na(west_and_rest)) |>
  mutate(time = factor(time)) |>
  mutate(world_6region = reorder(world_6region, dollars_per_day, FUN = median)) |>
  ggplot(aes(world_6region, dollars_per_day, fill = time)) +
  geom_boxplot() +
  geom_beeswarm(dodge.width = 0.8, size = 1.5, cex = 1.5, alpha = 0.7) +
  scale_y_continuous(trans = "log2") +
  xlab("") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle=90, hjust=1))
```



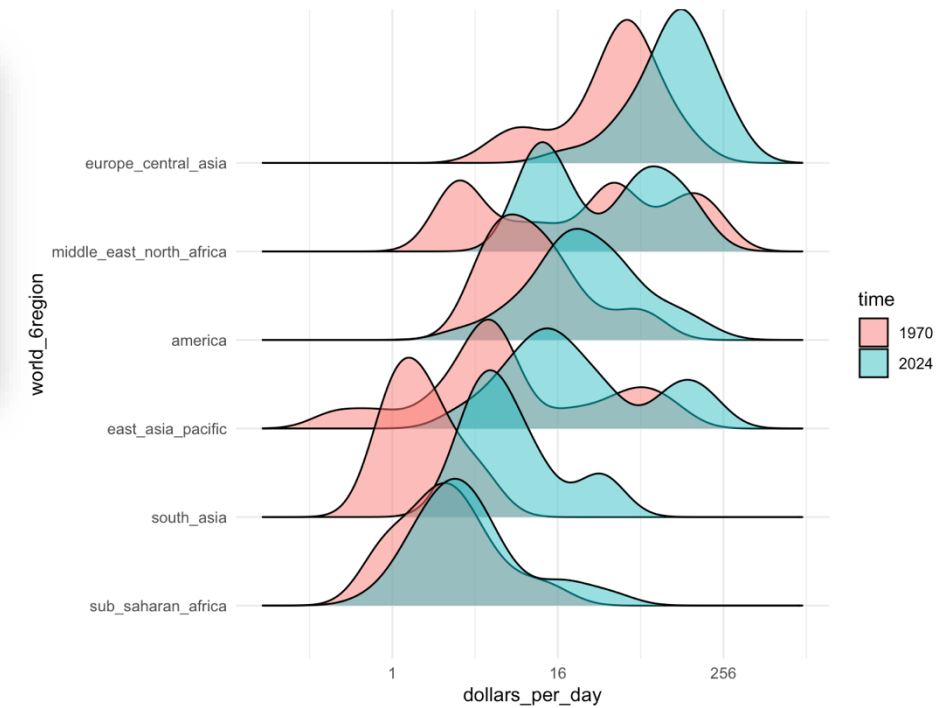
1970 versus 2024

Use ridge plot to show the density differences:

```
combined_df |>
  filter(time %in% c(past_year, present_year),
         name %in% country_list,
         !is.na(west_and_rest)) |>
  mutate(time = factor(time)) |>
  mutate(world_6region = reorder(world_6region, dollars_per_day, FUN = median)) |>
  ggplot(aes(dollars_per_day, world_6region, fill = time)) +
  scale_x_continuous(trans = "log2") +
  geom_density_ridges(alpha=0.5) +
  theme_minimal()
```

Takeaway from the plot:

- There still exists substantial income disparities, especially between Sub-Saharan Africa and the rest of the world.
- East Asia and South Asia have improved significantly
- The bi-modal distribution (i.e., dichotomy of west/rest) is dissipating





In this lecture

- Case study: fertility rate
- Case study: income
- Take-home question: What are some good ways to visualize **income distribution** across **regions** across continuous/multiple **time** points?